Diff: `ym-final` (29pp, Budden–Petron) vs the 160pp monograph (Petron)

Two proofs of the SU(2) Yang–Mills mass gap, not two drafts of one. Shared target and budget skeleton $\hat{\sigma}_\infty = \sigma_0 - \sum_k \eta_k$; divergent mechanism, scale, and bundling.

Architecture

	160pp monograph	**29pp `ym-final`**
Mass mechanism	Wilsonian RG + Kotecký–Preiss polymer/cluster expansion	Bałaban block-spin trajectory + Schwinger mechanism (BSE eigenvalue $\to 1$)
Inductive engine	sheaf-theoretic step; typed 2-cells Θ_k over a Gribov atlas	budget identity + Combes–Thomas decay $\to$ affine gap-map fixed point
Residual	5 conjectures + named analytic bundles (continuum measure, cluster bridge)	1 conjecture (`massgen`) + 2 cited published theorems (Bałaban, OS)
Constant	$m_0 \geq 999/1000$ pilot (g_0=0.30), fiducial-robust	reduction to $m < m_c$: m=0.50(7) vs $m_c \in [0.61, 0.70]$ GeV
Lean scale	67 thm / 43 def	6 closure modules + reduction backbone

Where the 160pp is stronger [HIGH]

- **Constructive finite layer.** Layer-1 $\mathbb{Q}'$-rational witnesses gate on `propext` *alone* — strictly tighter than our {`propext`, `Classical.choice`, `Quot.sound`}. Their finite arithmetic is choice-free; ours is not.
- **Robustness study.** Tornado + 5000-draw bootstrap over four fiducials ($c_\lambda, \mu_{\text{plaq}}, c_{\text{polymer}}, A_{\min}$); 5th-percentile $m_0 \geq 994/1000$. We have input-box scanning, nothing this systematic.
- **Self-containment.** Reproduces the Jaffe–Witten statement; $\sim$30 Python modules, $\geq$500 unit tests; fewer external dependencies.

Where `ym-final` is stronger [HIGH]

- **Honest bundling.** Our hard analysis is named and cited as *published* theorems (Bałaban CMP 1987–89, OS 1973/75). The monograph's Layer-2 continuum/OS/spectral results are Lean *type signatures assuming abstract analytic bundles* — the closure-critical analysis is assumed inside Lean, not proved. Its `propext`-only headline covers only the finite layer; it does not mean the gap is constructively proved. Our abstract states this boundary explicitly.
- **Sharper residual.** We reduced the open input to a single inequality between two computed masses, $m < m_c$. The monograph carries 5 conjectures (bounded looseness, Gaussian sharpness, union-bound looseness, $\hat{\sigma}_\infty$ convergence, SU(3) pilot).
- **Journal-readiness.** 29pp Annals-form, official Jaffe–Witten formulation, frame-compliant abstract; vs a 160pp monograph.

Shared gap [HIGH]

Both bottom out at the **cluster $\to$ spectral-gap bridge** and **OS reconstruction passing the gap to the continuum uniformly in L.** Monograph: analytic bundles + conjecture. Ours: BSE conjecture + cited OS. Neither closes it unconditionally; the `propext`-only gate does not change this, since the critical analysis sits in the monograph's assumed Layer-2 bundles.

Net

Monograph: deeper, more self-contained constructive *pilot*; tighter finite axiom-gate; real robustness analysis. `ym-final`: leaner reduction; single explicit residual $m < m_c$; published-theorem accounting of analytic inputs instead of in-Lean bundle assumptions; journal framing. The two transferable wins from the monograph are the `propext`-only rational-witness discipline and the bootstrap robustness study.